

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VI- Semester of (BSc) (Microbiology)

Examination April 2019

Course code & Name MI 602: Concepts of Microbial Ecology and Diversity

Date: 01-05-2019 Day: Wednesday Time: 1:50 PM To 3:30 PM Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q - II Answer the following questions as directed		
Q- II (A)	Answer any TWO of the following (i) Explain microbial biofilms and significance with a neat labelled diagram. (ii) What is the causative pathogen of tuberculosis? Define its classification. Explain any specific staining method for its detection. (iii) Explain a) Shanon diversity index b) Simpsons index.	8
Q- II (B)	Answer any ONE of the following (i) How yeast cells can be distinguished from bacterial cells. (ii) Define a) Prions b) Viroids.	2
SECTION - II		
Q-III Answer the following questions as directed		
Q- III (A)	Answer any ONE of the following (i) Define culture independent method of studying microbial diversity. Explain any two culture independent methods. (ii) Explain any two molecular characteristics that provide robust analysis of microbial evolution.	4
Q- III (B)	Answer any THREE of the following (i) Why rRNA genes are excellent candidates for phylogenetic analysis. (ii) Which group of extremophiles are present at higher temperatures and give any survival strategies of these microbes in such extreme conditions. (iii) Short note on Phylogenetic trees and significance. (iv) Discuss a) <i>Enterobacteriales</i> and b) <i>Pseudomonadales</i> with example.	9
Q- III (C)	Answer any ONE of the following	2

	(i) Draw a neat labelled diagram of a) Winogradsky's column and b) Microbial mats. (ii) Define a) Barophiles b) Xerophiles.	
--	---	--